

Q1. Reverse a list

Not Submitted

Q2. Reverse a string

Not Submitted

Q3. Searching

CODE

```
#include<stdio.h>
int main()
{
    int n,key,low=0,found=0;
    printf("enter the size of array:\n");
    scanf("%d",&n);
    int arr[n],high=n-1;
    printf("enter %d sorted element\n",n);
    for(int i=0;i<n;i++)
        scanf("%d",&arr[i]);
    printf("enter the element to search:\n");
    scanf("%d",&key);
    while(low<high){
        int mid=(low+high)/2;
        if(arr[mid]==key){
            printf("element found at index %d\n",mid);
            found=1;
            break;
        }
        if(arr[mid]<key)
            low=mid+1;
        else
            high=mid-1;
    }
    if (!found)
        printf("element not found\n");
    return 0;
}
```

INPUT

```
6
10 20 30 40 50 60
40
```

OUTPUT

```
enter the size of array:
enter 6 sorted element
enter the element to search:
element found at index 3
```

Q4. Searching

Not Submitted

Q5. Array

CODE

```
#include<stdio.h>
int main()
{
    int n,key;
    printf("enter size of array:");
    scanf("%d",&n);
    int arr[n];
    printf("enter %d sorted element:",n);
    for(int i=0;i<n;i++)
        scanf("%d",&arr[i]);
    printf("enter element to search:");
    scanf("%d",&key);
    int low=0,high=n-1;
    while (low<high)
    {
        int mid=(low+high)/2;
        if (arr[mid]== key){
            printf("element found at index %d\n",mid);
            return 0;
        }else if (arr[mid]<key){
            low=mid+1;
        }else{
            high=mid-1;
        }
    }
    printf("element not found\n");
    return 0;
}
```

```
int main()
{
    int n,key;
    printf("enter size of array:");
    scanf("%d",&n);

INPUT      5
          20 20 30 40 50 00
          50

OUTPUT    enter size of array:enter 5 sorted element:enter element to search:element found at index 4
```

Q6. Merging

— Not Submitted —

Q7. Largest and Smallest element

CODE

```
#include<stdio.h>
int main()
{
    int n,i;
    printf("enter the number of element:");
    scanf("%d",&n);
    int arr[n];
    printf("enter the %d element:",n);
    for(i=0;i<n;i++)
    {
        scanf("%d",&arr[i]);
    }
    int largest=arr[0];
    int smallest=arr[0];
    for(i=0;i<n;i++)
    {
        if(arr[i]>largest)
            largest=arr[i];
        if(arr[i]<smallest)
            smallest=arr[i];
    }
    printf("largest element :%d\n",largest);
    printf("smallest element:%d\n",smallest);
    return 0;
}
```

```
INPUT      5
          8 9 3 3 5

OUTPUT    enter the number of element:enter the 5 element:largest element :9
          smallest element:3
```

Q8. Second Largest Element

— Not Submitted —

OUTPUT
enter the
number of
element :
enter the 5
element:
largest
element :9
Smallest
element

Q8.
Second
Largest
Element
- Not
Submitted

Q8. Second Largest Element

— Not Submitted —

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Q8. Occurrence of an element

CODE

```
#include<stdio.h>
int main()
{
    int n,i;
    printf("enter size of array:");
    scanf("%d",&n);
    int arr[n];
    printf("enter %d elements:",n);
    for(int i=0;i<n;i++){
        scanf("%d",&arr[i]);
    }
    printf("enter elements to find:");
    scanf("%d",&x);
    int first=-1,last=-1;
    for(int i=0;i<n;i++){
        if(arr[i]==x)
        {
            first=i;
            break;
        }
    }
    if (first!=-1){
        for(int i=n-1;i>=0;i--){
            if(arr[i]==x)
            {
                last=i;
                break;
            }
        }
        if(first==last)
        {
            printf("element %d not found in the array.\n",x);
        }
        else{
            printf("first occurrence of %d is at index %d\n",x,first);
            printf("last occurrence of %d is at index %d.\n",x,last);
        }
    }
    return 0;
}
```

INPUT
7
2 5 3 3 7 5 8
5

OUTPUT
enter size of array:enter 7 elements:enter elements to find:first occurrence of 5 is at index 1
last occurrence of 5 is at index 5.

Q10. Missing Element

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Q9. Occurrence of an elemenet

CODE

```
#include<stdio.h>
int main()

    int n, x;
    printf("enter size of array: ");
    scanf("%d", &n);
    int arr[n];
    printf("enter %d elements:", n);
    for(int i=0; i<n; i++){
        scanf("%d", &arr[i]);

        printf("enter elements to find: ");
        scanf("%d", &x);
        int first = -1, last = -1;
        for(int i=0; i<n; i++){
            if(arr[i] == x)

                {
                    first=i; break;

                }

            if (First#-1){
                for(int i=n-1; i>=0; i--){
                    if(arr[i]==x)

                        last=i;
                        break;

                }

                if(First-1)

            }

            {

                printf("element %d not found in the array.\n", x);

            }
            }else{

                printf("first occurrence of %d is at index %d\n", x, first);
                printf("last occurrence of %d
                is at index %d.\n", x, last);
                return 0;

            }

        }

    }

    INPUT

    OUTPUT enter size of array - enter 7 elements - enter elements to find first occurrence of 5 is at index 1 last occurrence of 5
    is at index 5.

    2 5 3 5 7 5 8

    Q10. Missing Element
    - Not Submitted
```

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